KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2024
Department of Computer Science and Engineering
CSE 4127
Image Processing and Computer Vision

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Briefly describe the role of different components of an image processing system. (06) [CO1]
b) A lunar rover captured a 4-bit image of moon-surface under poor lighting condition, then it transmitted multiple 6×6 RLC patches. You are given an RLC image of a patch and a PDF. Your task is to adjust the pixel intensity according to the given PDF, then stretch the contrast of the output/obtained image to its maximum possible range. (20) [CO3]
RLC image: <3,3>, <4,2>, <3,3>, <6,4>, <5,4>,
<3,5>, <7,2>, <4,5>, <6,4>, <7,4>
PDF:

x	0	1	2	3	4	5	6	7	8	9
$f(x)$	0.06	0.07	0.1	0.15	0.2	0.15	0.1	0.07	0.05	0.05

c) Prove that 'convolution in time domain corresponds to multiplication in frequency domain'. (09) [CO2]

2. a) Explain the impact of wrong choice of order for contra-harmonic mean filter. (05) [CO2]
b) All the MRI images captured by an MRI machine contains periodic noises of different frequencies within the band (6 to 10). Design a butterworth filter to visualize the noise pattern. (10) [CO4]
c) Given a grayscale 4×4 image of $f(x,y)$ and a quantization matrix Q. Apply JPEG algorithm and compute the RMS error of compression-decompression. (20) [CO3]

$$f(x,y) = \begin{bmatrix} 52 & 55 & 61 & 66 \\ 70 & 61 & 64 & 73 \\ 63 & 59 & 55 & 90 \\ 67 & 73 & 78 & 85 \end{bmatrix} \quad Q = \begin{bmatrix} 16 & 11 & 10 & 16 \\ 12 & 12 & 14 & 19 \\ 14 & 13 & 16 & 24 \\ 14 & 17 & 22 & 29 \end{bmatrix}$$

3. a) Compute the Golomb code for the following representation: (10) [CO3]
i) $G_{12}(-15)$
ii) $G^2_{exp}(13)$
b) Apply pruning on the given binary image (Fig: 3(b)) using the structuring elements ($B1 \sim B8$) where- (15) [CO2]

$B1$=

x	0	0
1	1	0
x	0	0

$B5$=

1	0	0
0	1	0
0	0	0

$B2, B3$ & $B4$ are 90° rotated version of B_{i-1} from $B1$
$B6, B7$ & $B8$ are 90° rotated version of B_{i-1} from $B5$

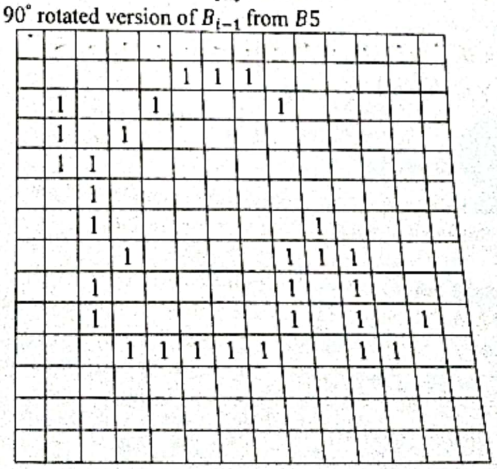


Fig: 3(b)

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c) Develop a mathematical model for degradation due to uniform linear motion. (10) [CO4]

4. a) Find the frequency domain filter $H(u,v)$ that works as a Laplacian filter in spatial domain, hence design a Wiener filter using $H(u,v)$ taking $SNR = 10$. (12) [CO3]
   b) Explain the procedure of morphological reconstruction with its application. (10) [CO1]
   c) A satellite captured an image of earth when one side of the earth is exposed to sunlight and other side remained dark, resulting a non-uniform illumination. Further, the memory of the satellite was lit by the cosmic rays causing an addition of very dense impulse noise to the stored image. What are the appropriate filters that can be used one after another to remove the impulse noises then to suppress the illumination? (13) [CO4]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Perform a full convolution on the following image using the given kernel. Be aware of the input and output indices. Use border replication padding while convolving. (10) [CO1]

	0	1	2	3
0	5	6	8	2
1	3	1	0	2
2	7	8	3	9
3	3	4	5	1

image

	-1	0	1
0	1	2	-2
1	-1	0	1

kernel

   b) Prove that convolution with a 2D separable filter can be accomplished by performing two one-dimensional convolutions. (09) [CO2]
   c) Suppose you want to segment the following image using normalized graph cut method proposed by Shi-Malik. Represent the image as a weighted graph considering 4-connected pixels and $\sigma_F = 4$; $\sigma_s = 1.5$, where symbols have their usual meaning. (16) [CO4]

8	4	9	25
5	11	22	28
9	2	27	30
29	27	28	31

6. a) Consider a grayscale image where foreground and background pixels are normally distributed with $\mu_f = 150, \mu_b = 50, \sigma_f = \sigma_b = 15$ and foreground pixels are one-third of total pixels. Find a minimum error threshold to separate the foreground from background. (09) [CO2]
   b) For the following binary image, compute first two invariant *Hu* moments. Equations given below are for your convenience, where symbols bear usual meanings. (15) [CO3]

y \ x	0	1	2	3	4
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	1	1	0
3	1	1	1	1	0
4	0	0	1	1	0

$$\emptyset_1 = \eta_{20} + \eta_{02}; \emptyset_2 = (\eta_{20} - \eta_{02})^2 + 4\eta_{11}^2$$

$$\eta_{pq} = \frac{\mu^{pq}}{\mu_{00}^{\gamma}}; \gamma = \frac{p+q}{2} + 1$$

$$\mu_{pq} = \sum\sum (x - \bar{x})^p (y - \bar{y})^q f(x,y); \bar{x} = \frac{m_{10}}{m_{00}}; \bar{y} = \frac{m_{01}}{m_{00}}$$

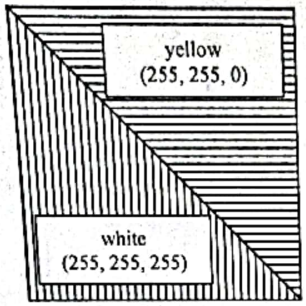
$$m_{pq} = \sum\sum x^p y^q f(x,y)$$

   c) Briefly explain Harris-Stephens corner detection method to find interest points from an image. Why is this method not covariant to image scaling? (11) [CO2]

7. a) What does it mean by homography transform of an image? Explain how can you estimate homography matrix using direct linear transform from two images. (12) [CO2]
   b) Following figure shows a RGB image with a white and a yellow region, where values in the parenthesis means RGB channel values. Compute edge magnitude and orientation at the diagonal line separating the regions using- (16) [CO3]
      (i) Di Zenzo-Cumani color edge operator.

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(ii) First convert the image to grayscale then apply monochromatic edge operator. Use simplest possible derivative operator.



c) State the pros and cons of histogram based texture descriptors with proper example. (07) [CO3]

8. a) How can you modify the Lucas-Kanade optical flow estimation method to handle reasonably large motion? Explain with necessary illustration. Do you need image warping? If so, why? (13) [CO3]

b) The following table shows some features extracted from some binary images along with their known classes. If you would like to use them as training set of your Naïve Bayes classifier, then, estimate the classifier parameters. Also, determine the class of the test sample (0.5, 0.42, 260, 0.59). (16) [CO4]

Form factor	Roundness	Compactness	Eccentricity	Class
0.455	0.327	276	0.587	Star
0.469	0.416	268	0.26	
0.48	0.34	262	0.64	
0.453	0.29	277	0.75	
0.78	0.66	161	0.46	Hexagram
0.79	0.58	159	0.66	
0.82	0.54	153	0.67	
0.84	0.56	148	0.62	

[Hint: Estimate μ_{ji}, σ_{ji} for features x_j and class c_i]

c) Explain how histogram intersection can be used as a distance measure for feature descriptor matching. (06) [CO1]

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